INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

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Course: **Computational Neuroscience (BT6270)** Date: October 12, 2021

Note: pplane8 is used of drawing phase plots

Poblem Description

Simulate the two variable **FitzHugh-Nagumo** neuron model using the following equations:

$$\frac{dv}{dt} = f(v) - w + I_{ext}, \frac{dw}{dt} = bv - rw$$

where $f(v) = v(a-v)(v-1)$. The parameters can be initialised as $a = 0.5$, $b = 0.1$ and $r = 0.1$. To solve the above differential equations Euler Integration method can be used.

Case 1: $\mathrm{I}_{ext} = 0\ \mu A$

1. **Draw a phase plot**.

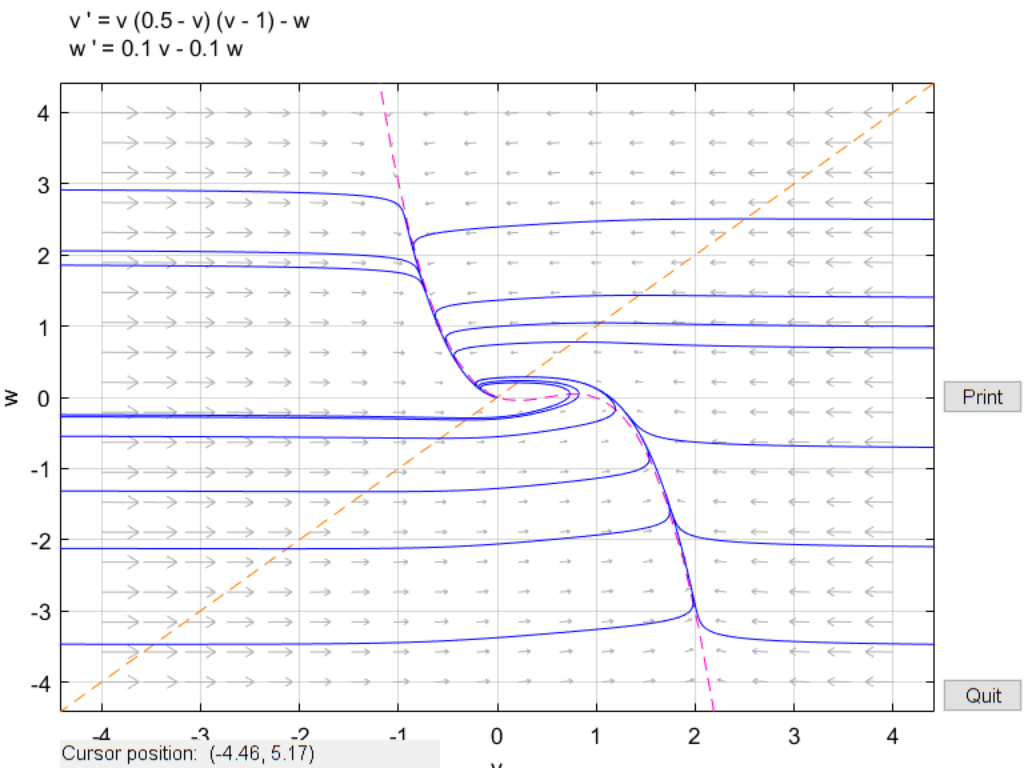


Figure 1: Phase plot for $\mathrm{I}_{ext} = 0\ \mu A$

2. **Plot $v(t)$ vs t, $w(t)$ vs t and also show the trajectory on the phase plane for the both sub-cases:**

   - $v(0) < a$ and $w(0) = 0$

     Let $v(0) = 0.2$

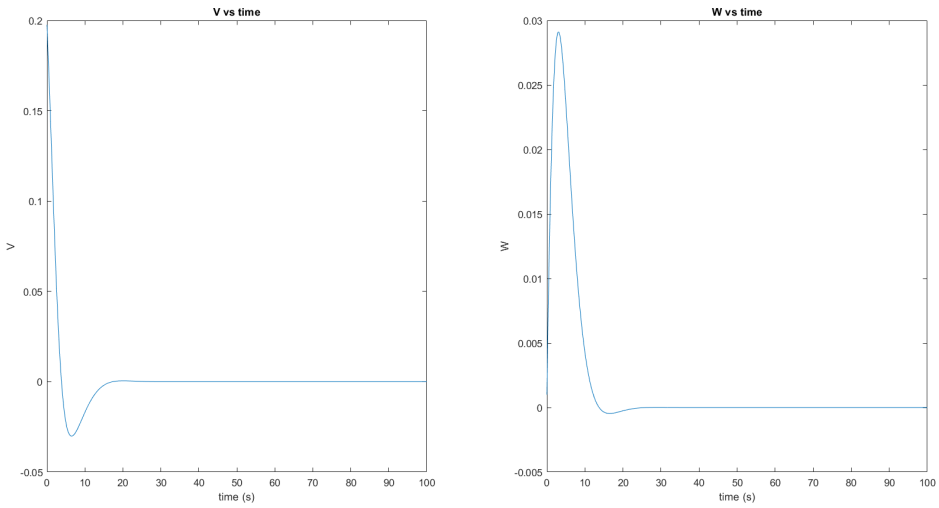


Figure 2: Plot of v vs t and w vs t for $v(0) = 0.2$ and $w(0) = 0$

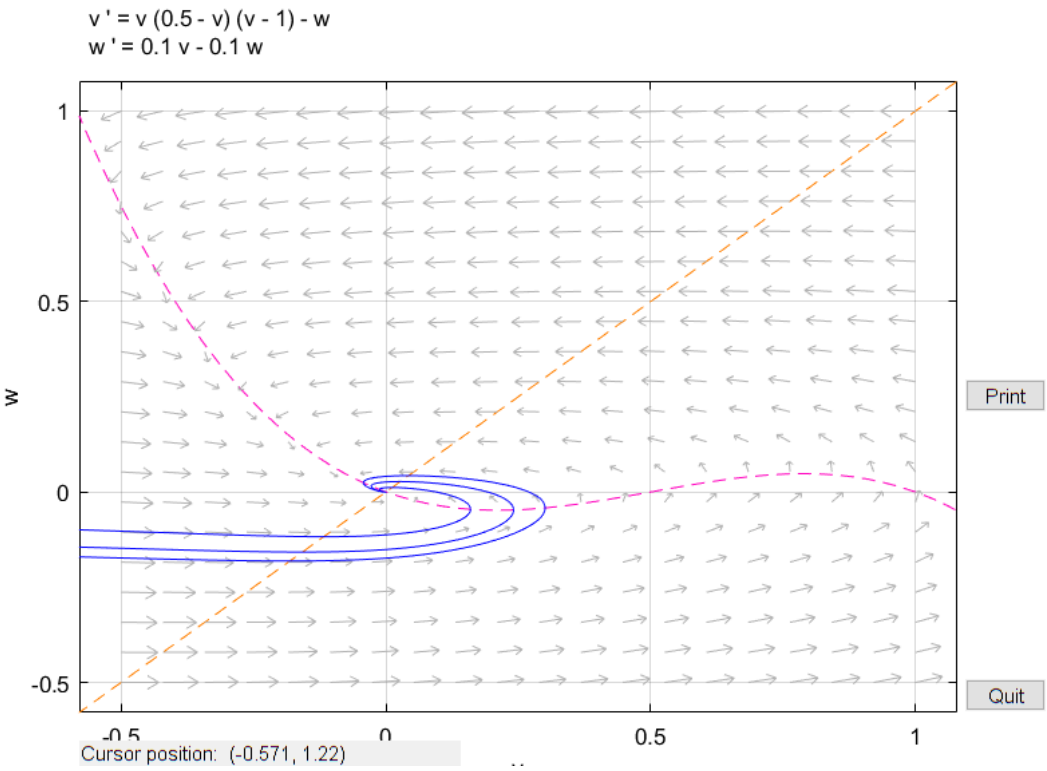


Figure 3: Phase Plane for $I_{ext} = 0$, $v(0) = 0.2$ and $w(0) = 0$

- $v(0) > a$ and $w(0) = 0$

  Let $v(0) = 0.6$

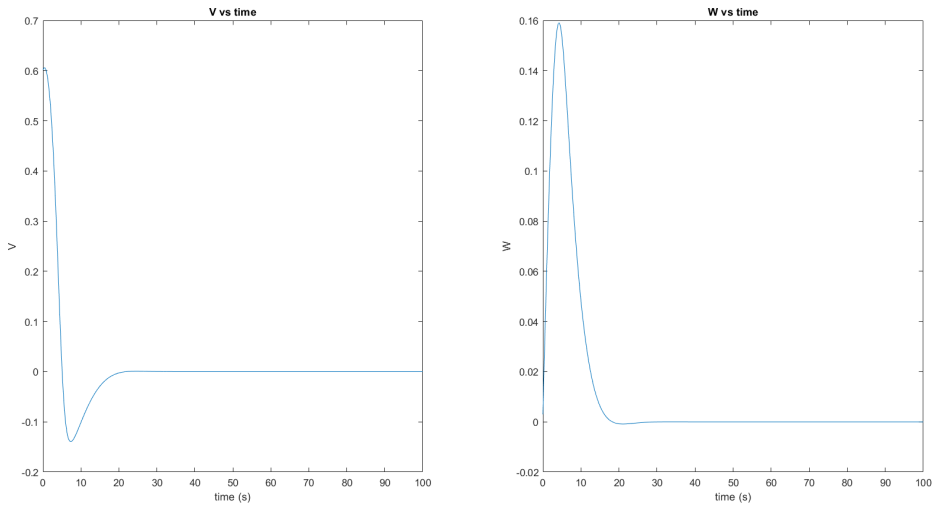


Figure 4: Plot of v vs t and w vs t for $v(0) = 0.6$ and $w(0) = 0$

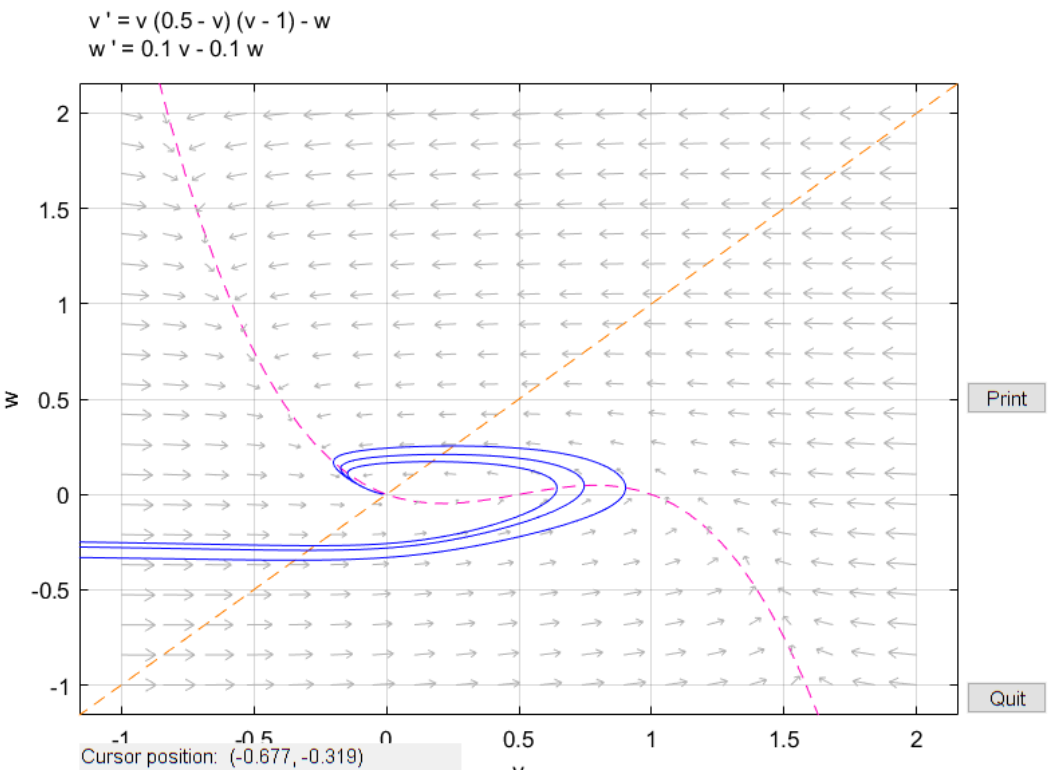


Figure 5: Phase Plane for $I_{ext} = 0$, $v(0) = 0.6$ and $w(0) = 0$

Case 2: $\mathrm{I}_{ext} = 0.5\ \mu A$ Limit Cycles

Choose some current value between I_1 and I_2 where it exhibits oscillations. Find the values of I_1 and I_2.

To calculate I_1 and I_2, we need maxima and minima of $f(v)$.
$\frac{df(v)}{dv} = 0$, therefore we get $v_{min} = 0.2113$ and $v_{max} = 0.7887$ as point of minimum and maximum value of $f(v)$.
For intersection with w-nullcline, points should lie on the other nullcline, substituting v_{min} and v_{max} in

$$v = v(0.5 - v)(1 - v) + \mathrm{I}_{ext}$$

will yield the following:

$$\mathrm{I}_1 = 0.2594\ \mu A \quad \mathrm{I}_2 = 0.7405\ \mu A$$

1. **Draw a Phase Plot for some sample value of I_{ext}.**

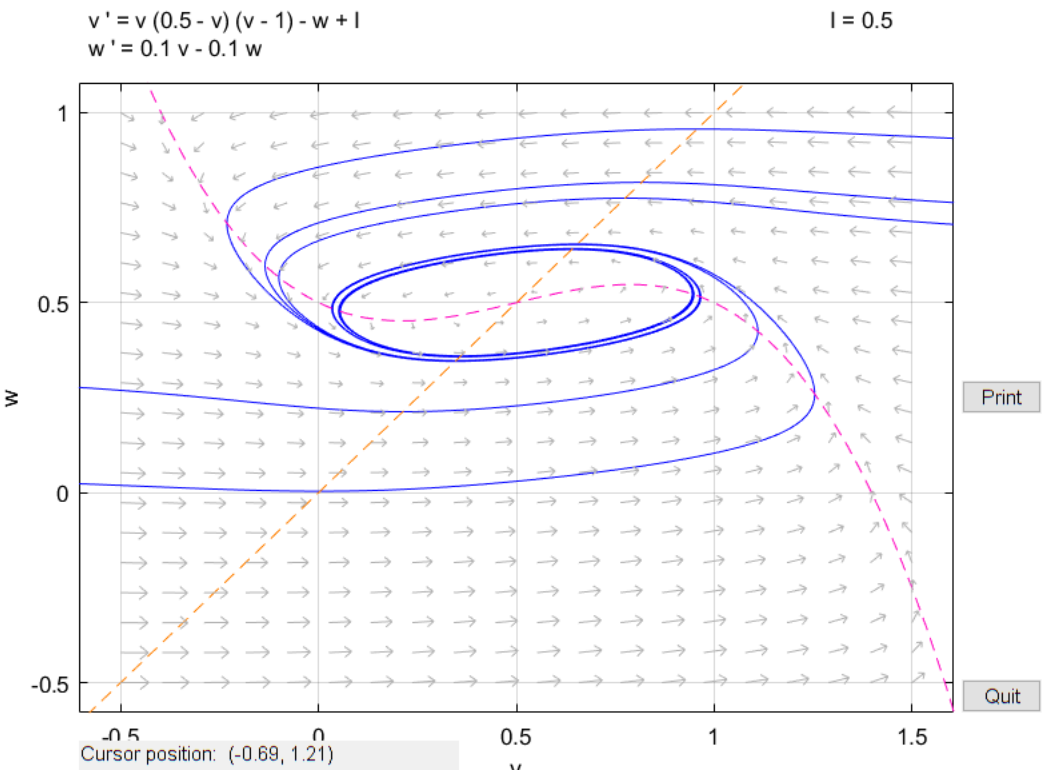


Figure 6: Phase plot for $\mathrm{I}_{ext} = 0.5\ \mu A$

2. **Show that the fixed point is unstable i.e., for a small perturbation there is no return to the fixed point (show the trajectory on the phase plane) – also show limit cycle on the phase plane.**

   As the value of external current increases and falls in between I1 and I2, the w-nullcline

intersects the v-nullcline in at a point where v-nullcline has a positive slope. For this case, there is only a single fixed point-(0.5,0.5) for which the following relations hold true:

$$f'(v) < \frac{b}{r} = 1, \Delta > 0$$

$$\tau = f'(v) > 0$$

Hence the given point is unstable.

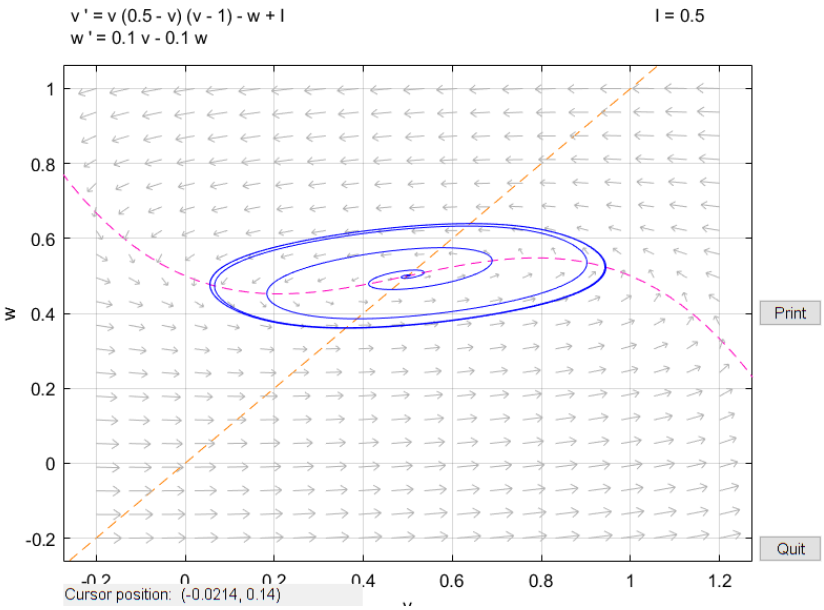


Figure 7: Small perturbations leading to a limit cycle

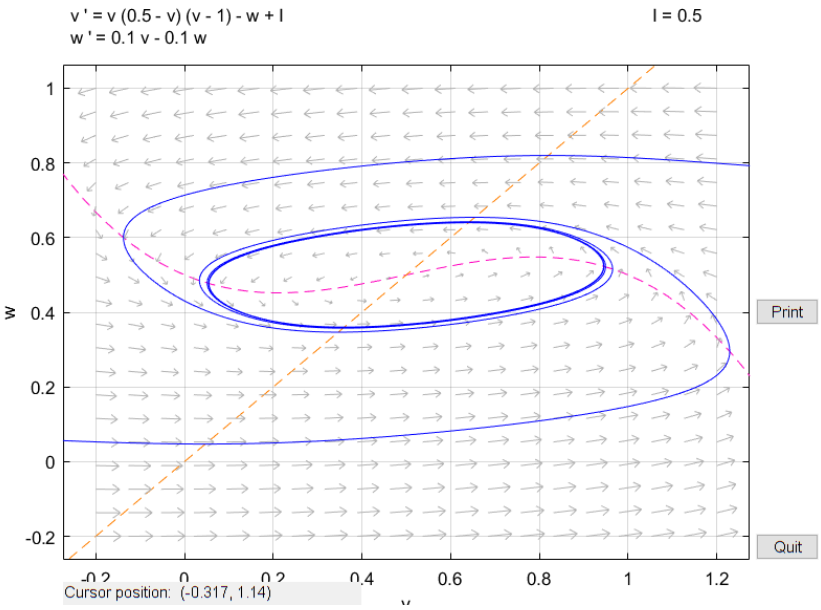


Figure 8: Limit Cycle on the Phase plane

3. **Plot $v(t)$ vs t and $w(t)$ vs t.**

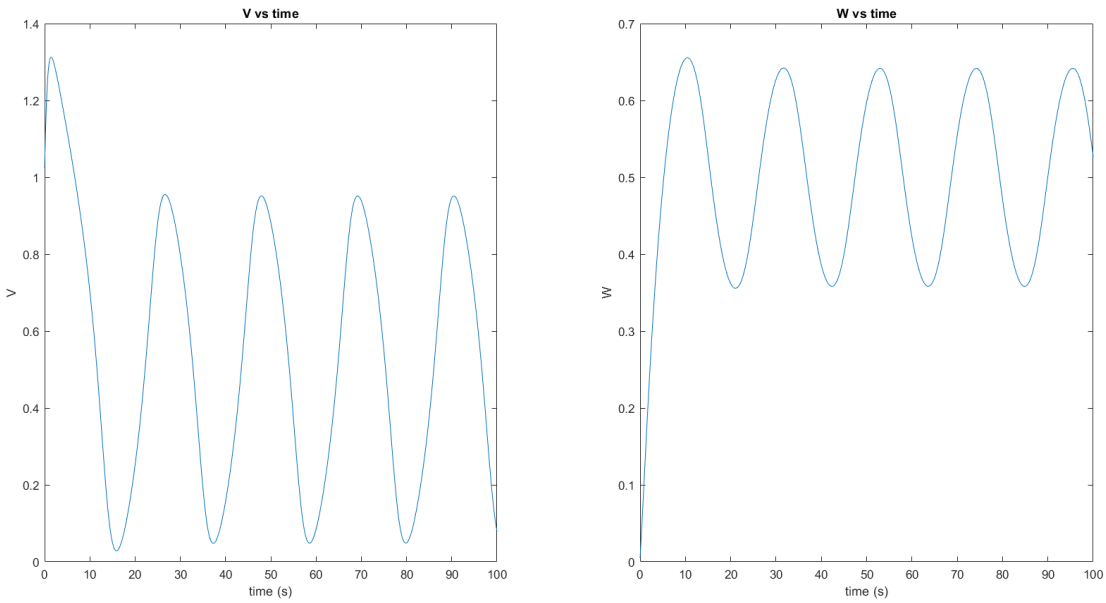


Figure 9: Plot of v vs t and w vs t for $v(0) = 1$ and $w(0) = 0$

Case 3: $I_{ext} = 1.0\ \mu A$ [Depolarisation]

1. **Draw a Phase Plot for some sample value of I_{ext}.**

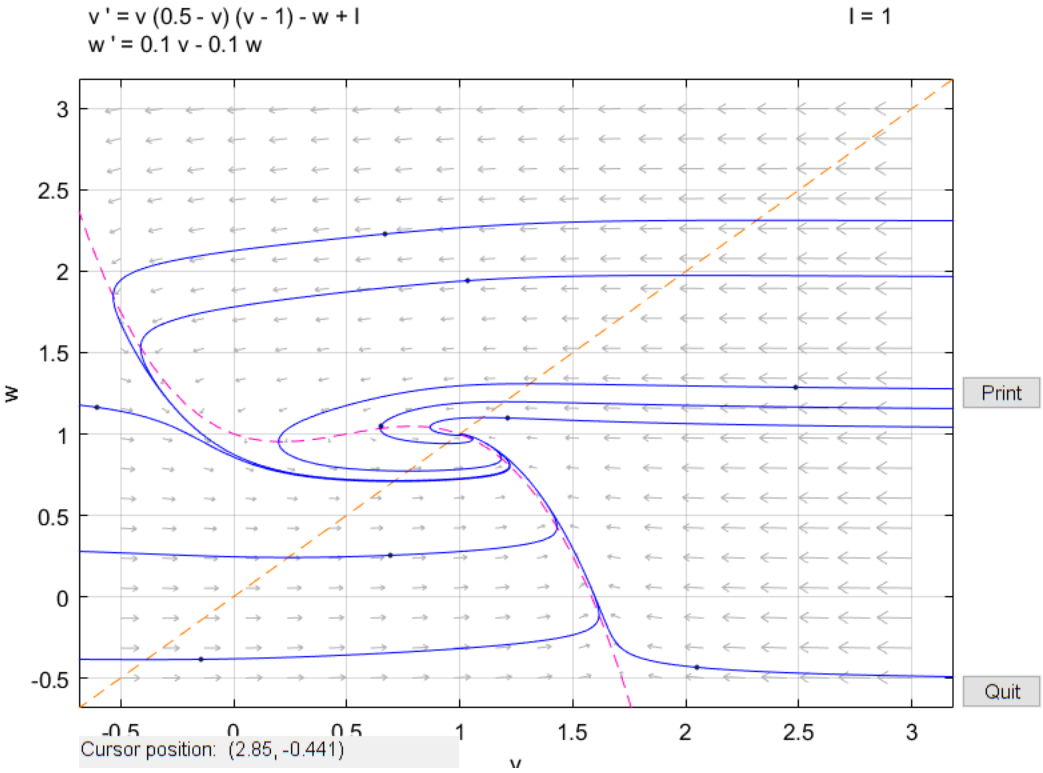


Figure 10: Phase plot for $I_{ext} = 1\ \mu A$

2. **Show that the fixed point is stable i.e., for a small perturbation there is a return to the fixed point (show the trajectory on the phase plane).**

   Upon further increment in I_{ext} the two nullclines intersect at the right branch of the v-nullcline, where the slope is negative. In this case there exist a fixed point - (1.0,1.0) for which

$$f'(v) < 0 < \frac{b}{r}, \Delta > 0$$

$$\tau = f'(v) < 0$$

   Therefore, the fixed point is stable. As seen in the Fig. 11

3. **Plot $v(t)$ vs t and $w(t)$ vs t.**

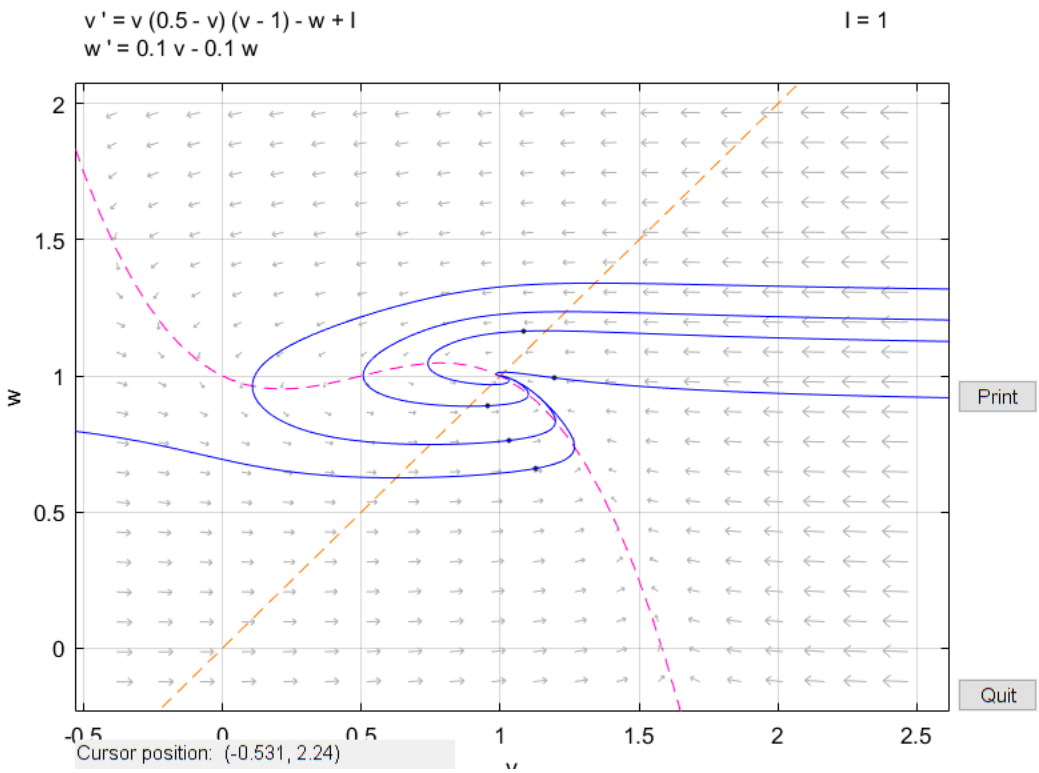


Figure 11: Small perturbations leading to depolarisation

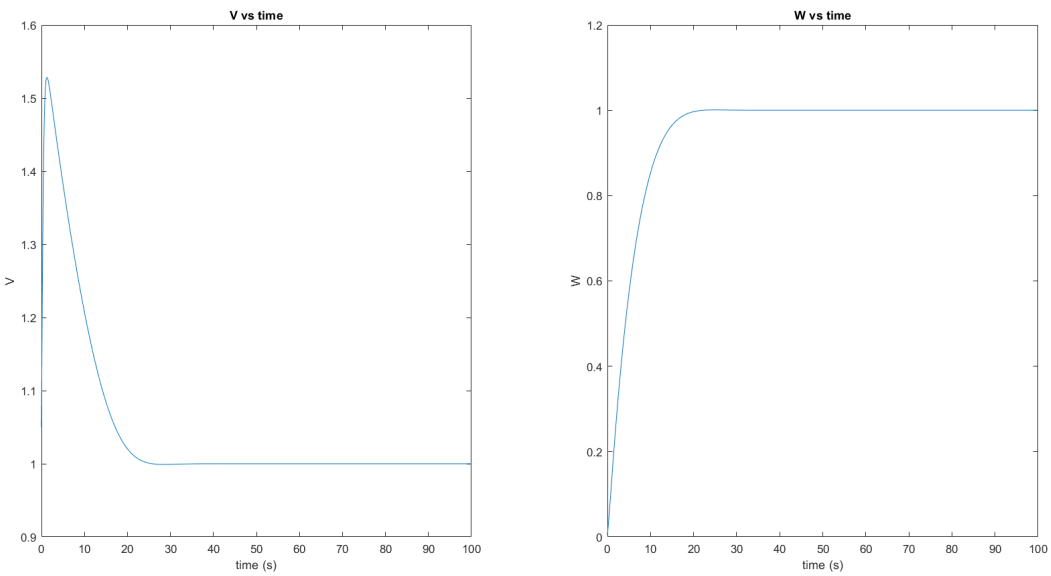
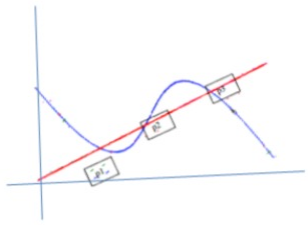


Figure 12: Plot of v vs t and w vs t for $v(0) = 1$ and $w(0) = 0$

Case 4: $\mathrm{I}_{ext} = 0\ \mu A$ [Bi-stability]

Find suitable values of I_{ext} and $\frac{b}{r}$ such that the graph looks as phase plot shown as below

1. **Redraw the Phase plot.**



For this case we assume that $I_{ext} = 0\ A$ and $a = 0.5$. On plotting both the nullclines, we can observe say that upon reducing the slope w-nullcline below 0.06098, causes it to intersect v-nullcline at 3 different points.

$$\frac{b}{r} < 0.069098$$

In our case let $r = 0.5$ and $b = 0.02$

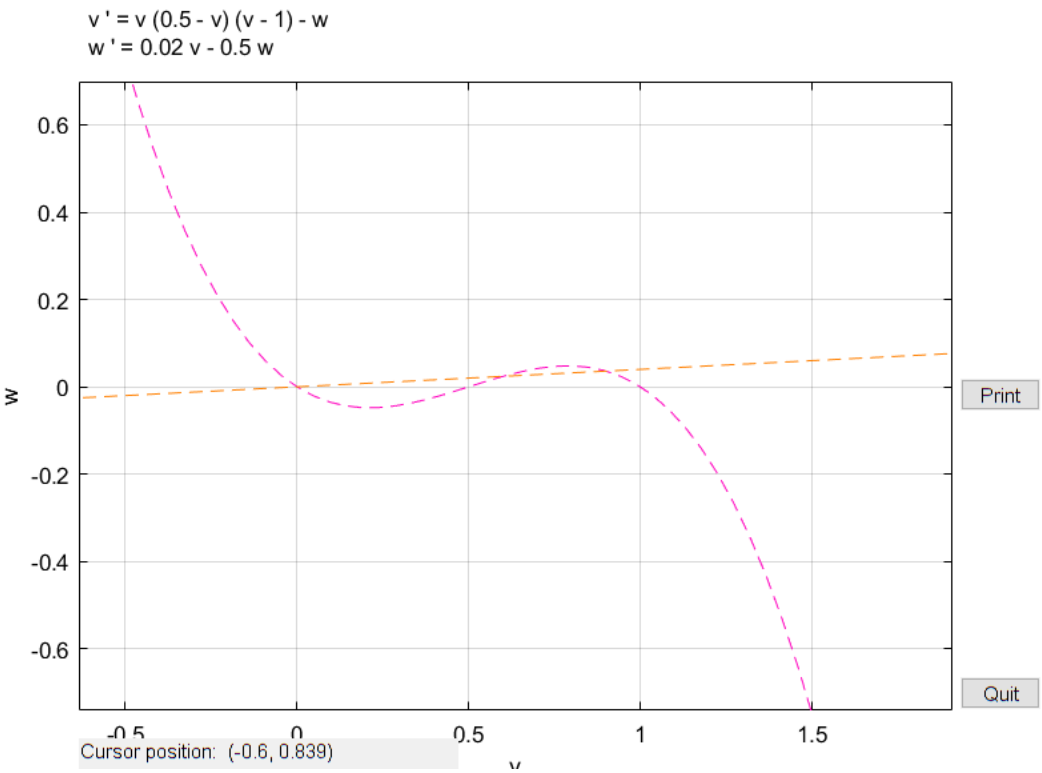


Figure 13: Phase plot for $I_{ext} = 0$, $a = 0.5$, $b = 0.02$, $r = 0.5$

2. **Stability of points P1, P2 and P3.**

   - **Point P1**
     $f'(v) < 0 < \frac{b}{r}$
     $\Delta > 0$
     $\tau = f'(v) < 0$
     Thereofore, point P1 is stable

- **Point P2**
  $f'(v) > \frac{b}{r}$
  $\Delta, 0$
  $\tau = f'(v) > 0$
  Thereofore, point P2 is a saddle point
- **Point P3**
  $f'(v) < 0 < \frac{b}{r}$
  $\Delta > 0$
  $\tau = f'(v) < 0$
  Thereofore, point P3 is stable

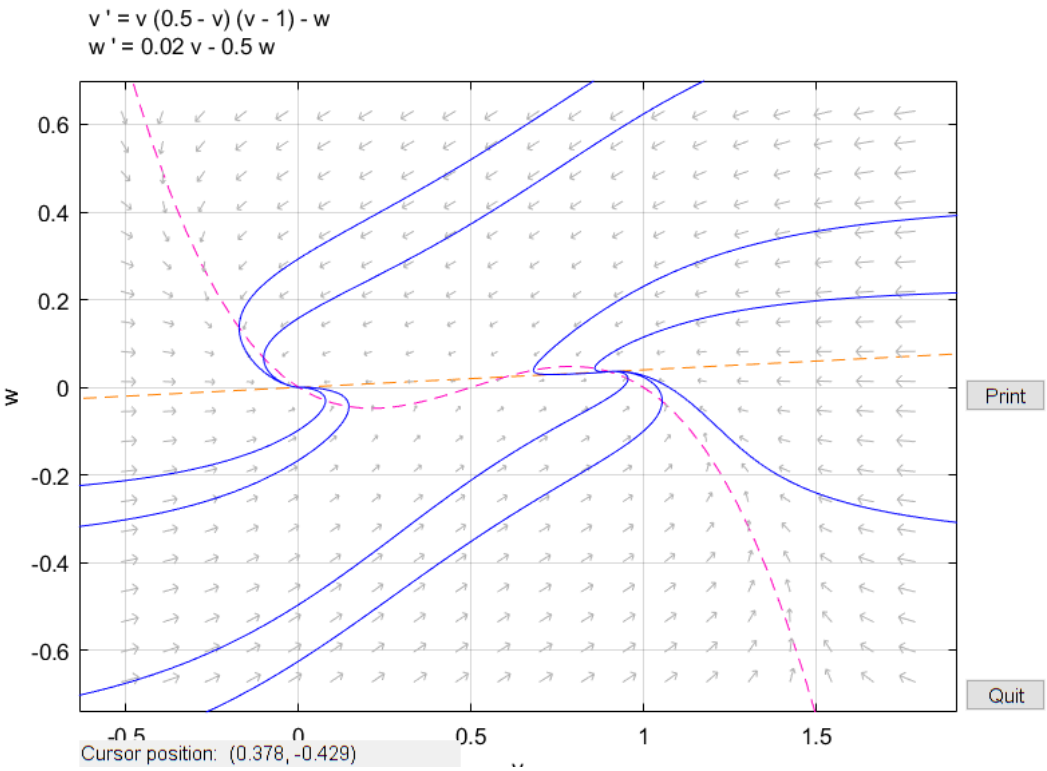


Figure 14: Small perturbations near P1 and P3

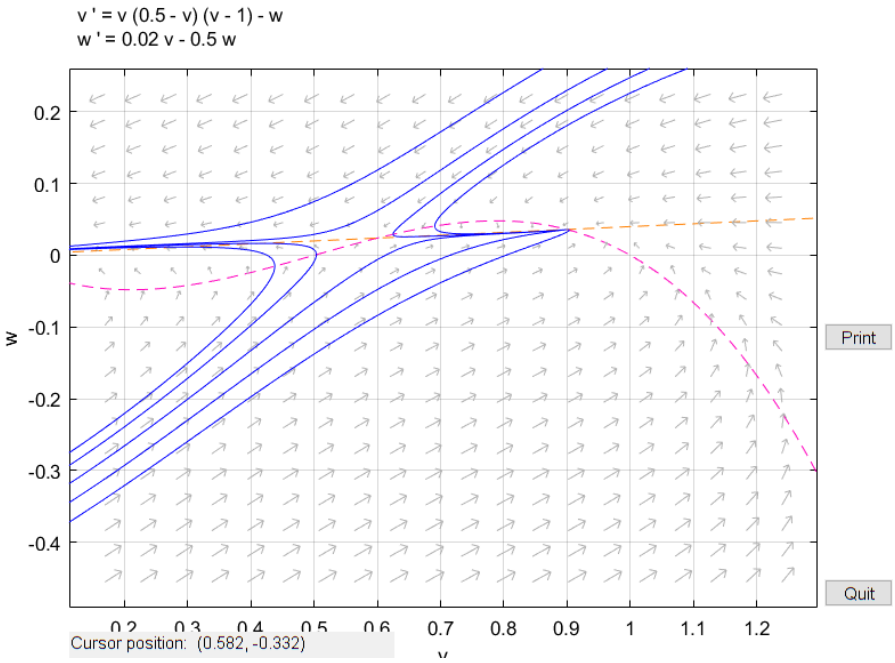


Figure 15: Small perturbations near P2

3. **Plot $v(t)$ vs t and $w(t)$ vs t.**

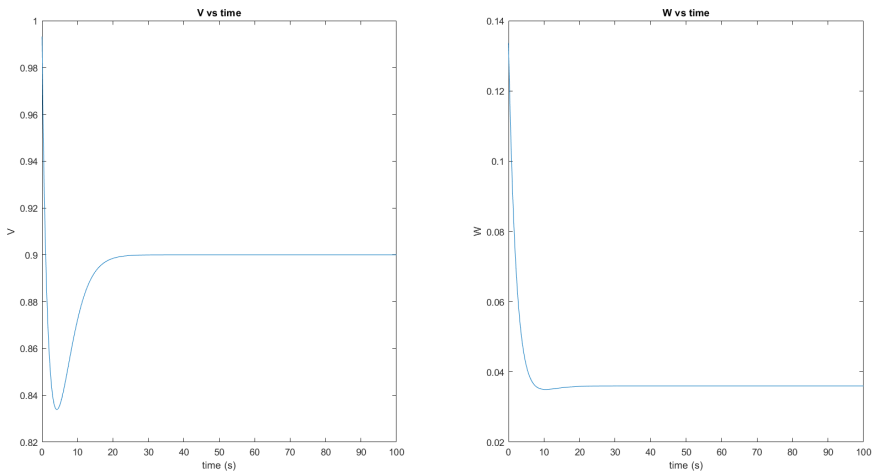


Figure 16: Plot of v vs t and w vs t for $\mathrm{I}_{ext} = 0$ and $\frac{b}{r} = 0.04$